

Qatar-1b

R-2688

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_110909 · created 2026-08-31T01:09:31+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2455813.7755 +/- 0.0004 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.02 +/- 0.056
Transit depth (Rp/Rs)^2	0.038 +/- 0.22 [%]
Orbital Inclination (inc)	86.82 +/- 2.9
Airmass coefficient 1 (a1)	0.0263993 +/- 2.7e-06
Airmass coefficient 2 (a2)	2.56 +/- 0.43
Scatter in the residuals of the lightcurve fit is	0.035 %
Best Comparison Star	#3 - [91, 28]
Optimal Aperture	32.62
Optimal Annulus	133.67
Transit Duration (day)	0.0132 +/- 0.005

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.02 ± 0.056 vs 0.1463 (deviation 2.26σ)
Duration fit vs published	0.0132 d vs 0.0692 d (deviation 11.19σ)
Mid-time O-C	-0.5 min
Depth SNR	0.4
Residual scatter	0.035 %

Light curve

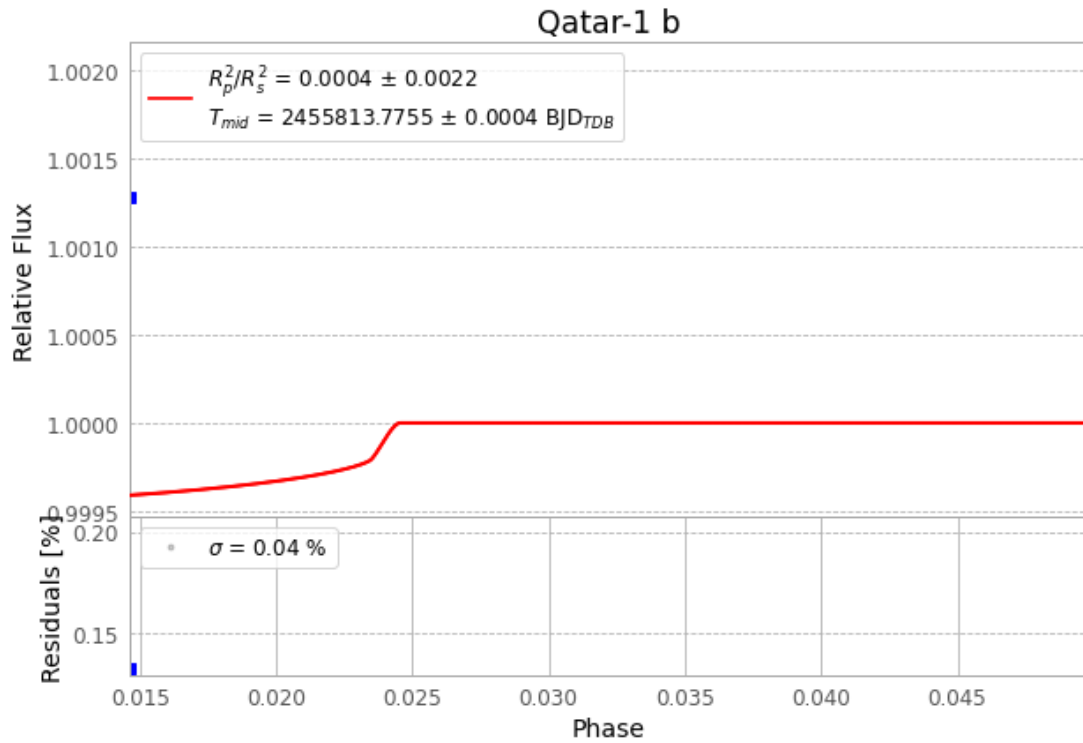


Figure 1 — FinalLightCurve_Qatar-1 b_2011-09-08.png (SHA-256 09fa5aa0a420c30e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [273, 288], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 02631e6098942f9c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

62 FITS frames (41 MB), Qatar-1110909051812.FITS → Qatar-1110909082712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-110909.log (1885cb7e791d...), FinalLightCurve_Qatar-1 b_2011-09-08.png (09fa5aa0a420...), FinalLightCurve_Qatar-1 b_2011-09-08.csv (1b34799540c3...)

Compute resources

Wall time 811.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-31 01:09Z.